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E-440 SUBSTATION PROTECTION RELAYING

1. SCOPE OF WORK

The work shall include the design, manufacturing, supply, factory testing, delivery to site, installation, commissioning and testing of the protective relaying requirements and the associated auxiliary relays for substation in accordance with Part 4, Subsections 1.2 Summary of work and the bid drawings.

The Contractor shall integrate the protective relaying system supplied by them to the existing substation protective relaying system such as to perform properly and effectively. Re-coordination and resetting shall be covered by this Specification and included in the Contractor scope of works.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4 Subsection 2.5 Codes and Standards and, the following specific standards.

2.1 Indonesian National Standards

PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60255 : Electrical relays

2.3 Institute of Electrical and Electronics Engineers

ANSI/IEEE C37.90 : Relays and Relay System associated with electric Power apparatus

3. OPERATING AND DESIGN CONDITIONS

The substation protection relaying system shall be designed for operation in the tropical climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for the meteorological, seismic and environmental conditions of the site.

4. DETAILED REQUIREMENTS

4.1 General

The substation protective relaying system shall provide high speed protection to all equipment with reliability, selectivity and security suitable for the 500 kV substations.

Operation of every protection system shall trip both trip coils of all the breakers necessary to isolate the fault. All the protective relay operations shall initiate a hand reset mechanically-latching type tripping relay which shall trip all the breakers necessary to isolate the fault and shall also interlock the closing circuits of the breakers to prevent their closing until the tripping relay is reset. Each tripping relay shall be provided with a target to indicate its position (red for trip and green for reset) and shall be reset-able from the front of the panel. Sufficient extra contacts shall be provided for alarm and interlocks.

Sequential time-delayed tripping shall only be permitted in the following cases:

- (1) Breaker failure protection.
- (2) Short zone protection as specified in Clause 4.6.
- (3) Complete failure of the fiber optic carrier communication link.
- (4) Extremely high fault resistance.

Minimum requirements for voltage transformers and current transformers associated with the relaying system are given in Technical Specifications E-420 and E-421. The Contractor shall supply the transformers suitable for the correct performance of the relaying system.

The protective system shall remain stable during switching operations on the power system irrespective of whether or not the operations directly affect the element of the power system protected by the relay system.

The protective system shall remain un-operated following the normal correct discharge operation of one (1) or more surge arresters.

4.2 Performance Requirements

The protective relaying system shall meet the following performance requirements :

- (1) Except for the time delayed tripping permitted in Clause 4.1 (1) to (4), all protection schemes shall clear the faults in the following times:

- a. 500 kV Faults.....90 ms
- b. Transformer Faults.....90 ms

- (2) The breaker failure protection shall operate all the back-up breaker to isolate the fault within the following times.

500 kV breakers.....200<t<250 ms

4.3 Bus Protection

Bus bar protections have been installed by the last project. The Contractor shall integrate their busbar protection auxiliaries and relaying scheme to the existing and shall re-coordinate and resetting the system properly and effectively. The Contractor is responsible to add any equipment as necessary for the proper functioning of the busbar protection.

The protection schemes shall be suitable for all out-of-zone, phase-to-phase and phase-to-earth fault currents up to the short circuit current rating of the circuit breakers, applied gradually or suddenly and irrespective of the distribution of current between the individual circuits.

The protection scheme shall not mal-operate as a result of faults in the associated secondary wiring when the primary circuit is carrying load current. Continuous or automatic cyclic supervision shall be provided for this purpose. Arrangements shall be provided to make the zone containing the faulty circuit inoperative. Where the operation of the supervision device is dependent upon the load current in the primary circuit, operation shall occur at currents of 25 amperes or 10% of the circuit rating whichever is the greater.

The following tests and isolation facilities shall be provided on the front of the protection panel.

- (1) Short circuiting and isolation of current transformers.
- (2) Isolation of tripping and individual zones.

The digital multi functional programmable relay shall be such that definite operation occurs for phase-to-phase and phase-to-ground short circuits during minimum plant conditions, irrespective of the number of circuits in service and the distribution of load and fault currents.

4.4 Branch Protection

A separate differential type protection for sections between two (2) breakers and the corresponding transformer switch shall be provided.

The differential type protection shall be coordinated with the protection for generator transformer and shall be similar to the bus protection described in Clause 4.3.

4.5 Transformer Protection

The Contractor shall provide and coordinate the appropriate relays to ensure a reliable transformer protection meeting the specified requirements.

The following are the minimum requirements:

(1) **Differential Protection**

- a. The differential relays shall be designed to provide a complete protection to the transformer. They shall protect the transformer against all phase-to-phase and phase-to-ground faults.
- b. The differential relays shall be properly restrained to prevent tripping for all out-of-zone faults and inrush current.
- c. The differential relays shall be designed to operate properly on all transformer tap-changer settings.

(2) **Restricted Earth Fault Protection**

The restricted earth fault protective relays shall protect the transformer windings as near as is practicable to the neutral points and at least 85% of all winding shall be protected.

(3) **Overcurrent Protection**

The Contractor shall coordinate the relay characteristics to provide a secure and coordinated protection system.

(4) **Transformer Protection Devices**

The transformer will be provided with Buchholz protective relays. They will also be provided with temperature tripping relays and possibly other protective relays. The Contractor shall connect the output of these relays in the overall transformer protection scheme and shall supply, install and connect

the necessary auxiliary relays.

4.6 Circuit Breaker Failure and Short Zone Protection

A circuit breaker failure protection scheme shall be provided for each 500 kV circuit breaker. In addition, a short zone protection shall be provided to protect the zone between the circuit breaker and the current transformers.

The circuit breakers failure protection shall operate when a fault has been correctly detected by a protective relay but the circuit breaker has failed to operate within a predetermined time. On operation of a circuit breaker failure protection, all circuit breakers connected to the failed circuit breakers including in the existing substation shall be tripped (both trip coils) to isolate the fault.

The circuit breakers failure protection shall be designed to provide a reliable back-up system. The dropout time of the current sensing elements shall be as low as practicable to provide the maximum-security margin. The time delay shall be selected to ensure that the main protection has sufficient time to operate and the fault is cleared within the specified time.

Each circuit breaker failure protection and each short zone protection shall be provided with a separate tripping relay. Combining these tripping relay with the bus protection tripping relay or other tripping relays is not permitted. The short zone protection may share other circuit breaker failure protection scheme component subject to the Owner's review.

The short zone protection shall operate when a circuit breaker has opened but the corresponding current sensing element still detects at current flow. The timer setting shall be coordinated to provide a speedy yet secure protection.

4.7 Relay Construction and Mounting

The relay construction and mounting shall be as described in Technical Specification E-280 except as modified/supplemented below:

- (1) Digital multi functional programmable relays with modular construction shall be used.
- (2) The relays and associated cables shall be adequately shielded against surges.
- (3) The relays shall be designed to prevent mal-operation due to the DC supply system surges.

- (4) The relays shall be provided with adequate number of indications to facilitate the post-fault identification of the faulted phase and zone (s). Sufficient key items from the line relays shall be connected to the event recorder to permit a thorough post-fault analysis.
- (5) Wherever practicable, the design of the relay system shall be based on the “fail-safe” principle. Loss of supply or a failed circuit component shall not cause an incorrect equipment operation.

5. PERFORMANCE TESTING AND GUARANTEES

The relay system performance testing and guarantees shall be as described in Technical Specification E-250.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control for the substation protection relaying shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedure specified shall included, but not be limited to, the followings:

- (1) All routine tests and specified tests specified.
- (2) All proposed site tests to verify the relaying system performance.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Report of routine tests and any other specified tests performed at the factory.
- (2) Report of all the site tests to verify the relay performance.
- (3) Relay setting calculations.
- (4) All other documentation required by applicable codes and standards.

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